Supplementary Materials

On the Automated Annotation of Interaction Mechanisms at Protein–Protein Interfaces

Human Annotation Guidelines and G250 Mutation–PDB Index

1. Human Annotation Guidelines

The following are the laboratory's original working instructions for identifying and documenting mutation-associated interactions in structural biology papers. The opening 1CSE entry is an illustrative annotation template, not a G250 dataset record. The guidelines include additional labels beyond the six interaction classes evaluated in the accompanying paper because they describe the broader manual annotation process. The instructions and example quotations below are preserved as supplied, including transcription artifacts in the source excerpts.

1CSE: COMPLEX FORMED BETWEEN SUBTILISIN CARLSBERG AND EGLIN C, AN ELASTASE INHIBITOR

Overview:

-- add a brief description of the function of the complex. Usually you can cut and paste from the abstract. Example:

Standard mechanism protein inhibitors of serine proteinases share a common mechanism of interaction with their cognate enzymes. The P1 residue of the inhibitor interacts with the enzyme in a substrate-like manner. Its side chain becomes imbedded in the S1 cavity of the enzyme. The nature of P1, the primary specificity residue, greatly affects the strength and specificity of the enzyme inhibitor association. In canonical inhibitors, residues P4-P2′(P3′), where P1-P1′ is the reactive site, share a common main chain conformation that does not change on complex formation. The remainder of the inhibitor's structure, the scaffolding, is not always common. Instead, there are at least 20 inhibitor families, each with a different scaffolding.

Chains:

-- indicate what proteins are which chains in the PDB file. Example.

A: Subtilisin Carlsberg

B: Eglin C

List the involved mutants:

AI123B (bold the mutant name. Use the Skempi format)

[mechanism] (state what mechanism affects the mutation)

<add the quote from the paper>

-—<cite the paper filename, so we can find the quote again>

AI123B (bold the mutant name. Use the Skempi format)

[mechanism] (state what mechanism affects the mutation)

<add the quote from the paper>

-—<cite the paper filename, so we can find the quote again>

Guidelines for mechanisms (mechanisms appear in :

- When starting a protein, **copy** all the papers related to the PDB file in the folder the folder called "[downloaded] all papers" to the subfolder for the pdb file. All papers start with their pdb code, so it's easy to find them. Make copies so we dont lose this centralized resource. Infinite google drive space thanks to Lehigh.
  - The structure paper is always named (pdbcode)-(author).pdf, whereas the experimentation papers from Skempi are always (pdbcode)-(number).pdf, to distinguish them
- The structure paper is almost always the best resource for the mechanism information about each amino acid
- When searching for an amino acid, search only for the number. I.E. if you are searching for the mutation AA123B, search for "123". This will give you a bunch of garbage to skip (e.g. page numbers, citations, etc) but it guarantees that you don't miss any text relating to the amino acid.

Mechanisms: [listed in square brackets]

[not documented]

-- This occurs. It means there are probably other papers about the mechanism that are not in the

research we have, but we are drawing the line at the downloaded papers.

-- This label indicates the amino acid does not come up when you search for the amino acid in all the papers from the papers folder.

-- This can also happen if the amino acid is mentioned, but only in terms of the affinity experiment and not in terms of mechanisms. (e.g. alanine scanning)

[no mechanism]

--In such cases, the paper actually says that this amino acid does not affect binding or specificity. Example:

Mutants at .. **Gln381, do not affect the binding at all**. These contact side chains are located toward the periphery of the interface and contribute less to the total contacts. An alanine replacement of Gln381 appears to be isofunctional, as this residue interacts with ZipA/M185 through the innermost aliphatic portion of its side chain.

--1f47-mosyak2000.pdf

[low impact]

-- In such cases, the text labels the amino acid as having a mechanism but also asserts that it doesnt really affect affinity. Example:

hRI Trp263, Trp318 and Trp375, whose replacements produce G values of 1.0-1.5 kcal/mol, undergo 34-60 AÊ 2 reductions in hydrophobic ASA upon complex formation. Substitutions of Trp261 and Lys320 of hRI, and His84 and Glu108 of Ang barely alter af®nity; these residues form only van der Waals contacts and the changes in non-polar ASA attributable to them are quite small. In four instances (replacements of Arg31 and Arg32 on Ang, and Tyr150 and Ser289 on hRI), disruption of contacts that technically qualify as hydrogen bonds had little or no effect.

--1a4y-001.pdf

[hbond]

-- Use this when the paper mentions that the amino acid is involved in the protein-protein interface and that it forms a hydrogen bond.

-- To be used for an intermolecular hbond. See example:

The remaining hydrogen bonds involve sidechains of each protein and include, Im9 Ser50 with DNase Gln92 and Im9 Asp62 with DNase Asn72 (Figure 4(a)). Of these, only Ser50 is conserved (this residue is threonine in Im7 but serine in the other immunity proteins).

—-1emv-10.1006@jmbi.2000.3945.pdf

[hbond] [salt bridge]

-- Use this when the paper mentions that the amino acid is involved in the protein-protein interface and that it forms a salt bridge. (some people say salt link). All salt bridges are also hydrogen bonds, so always use this next to [hbond]. See example:

Surrounding this hydrophobic pocket are 12 hydrogen bonds (Table 2 and Figure 4(a)), two being salt-bridges and three of which involve one charged partner. The two salt-links are formed between basic residues from the E9 DNase, Arg54 and Lys97, and acidic residues from the N and C-terminal ends of helix II from Im9, Glu30 and Glu41, respectively. Both salt-links likely contribute to the overall binding free energy, since alanine mutants of Glu30 and Glu41 of Im9 destabilise the complex by 1.4 kcal molÿ1 and 2.1 kcal molÿ1 respectively (Wallis et al., 1998). Sequence comparisons show that Im9 Glu30 and E9 DNase Arg54 are both conserved in their respective families, suggesting that this salt-bridge should be present in all the DNase-immunity protein complexes.

—-1emv-10.1006@jmbi.2000.3945.pdf

[hbond-internal]

-- Use this when a paper mentions that the hydrogen bond formed by this amino acid is intramolecular (e.g. internal to one chain) and that mutations of the amino acid breaks the internal hydrogen bond and interferes with binding.

Aspartic acids Asp370 and Asp373 are the only two side chains that do modulate binding, but are not involved in direct interaction with the ZipA/M185 molecule. Yet their functional importance is evident and can be accounted for by the structure. The subtle effect of the D370A mutant (3.2-fold decrease in binding) is correlated with a decrease in the number of van der Waals contacts that are seen near but outside of the contact interface. Those involve hydrogen bonds between the charged Asp370 and ZipA/M185 backbone atoms through the bridging water molecules (not shown).

--1f47-mosyak2000.pdf

[hbond] [hbond-waterMediated]

-- Use this when the paper mentions that the amino acid is involved in the protein-protein interface and that it stabilizes a water molecule. This is by definition an hbond, so include that label. See example:

Although none of the ®ve conserved E9 DNase residues in the Im9 binding site contacts the immunity protein directly, a conserved asparagine residue (Asn75) from the DNase is involved in co-

ordinating a bur-ied water molecule (W2), which in turn hydrogen bonds the backbone of both Im9 (carbonyl group of Tyr54) and the DNase (amide nitrogen atom of Asn72).

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[Hydrophobic]

-- Use this when an amino acid forms a hydrophobic interface. Papers will talk about hydrophobic interfaces, hydrophobic pockets, etc, that the amino acid is part of. Note that these generally only occur with regard to a hydrophobic amino acid being mutated. See example:

Thirty-one residues (14 from the E9 DNase and 17 from Im9) are in contact across the interface. In the centre of the binding site is an aromatic stack-ing interaction between Phe86 of the E9 DNase domain and Tyr54 of the immunity protein (Figure 4). The Phe86-Tyr54 interaction is lined by hydrophobic residues, including Leu33, Val34, Val37, Ile53 and the aromatic ring of Tyr55 of the immunity protein, as well as the alkyl side-chains of Lys89, Lys97 and Val98 of the DNase domain.

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[electrostatic]

-- Use this when the amino acid forms an electrostatic interaction between charged amino acids. Only five amino acids can form such an interface: the negative residues (ASP, GLU) and the positive residues (ARG, LYS, HIS). The paper will always say charged attraction, electrostatic attraction, or something about charge. Note that Electrostatic interactions often happen at the same time as salt bridges, so you might often have both. A salt bridge is just a hydrogen bond and a simultaneous electrostatic interaction. So you might have all three labels.

The binding affinities from the charge swap experiments were grouped into the same categories as the alanine mutant affinities. Six of the mutants lost >1.0 kcal/mol of binding free energy. These include UbcH7 helix 1 mutants R5E, R6E, and K9E… The helix 1 mutants form inter-chain (R5 and R6) and intra-chain (K9) hydrogen bonds to backbone oxygen atoms.

-1c4z-000.pdf

[stacking]

-- stacking interactions occur between amino acids that form rings (mostly aromatics). They occur rarely, and when they occur the paper will say explicitly that a stacking interaction is occurring, such as the example below.

Thirty-one residues (14 from the E9 DNase and 17 from Im9) are in contact across the interface. In the centre of the binding site is an aromatic stack-ing interaction between Phe86 of the E9 DNase domain and Tyr54 of the immunity protein (Figure 4). The roles of the two immunity protein tyrosine residues (Tyr54 and Tyr55) at the interface have also been probed by mutagenesis. On substitution for alanine, Tyr54, which stacks against Phe86 of the DNase, has a signi®cant effect on DNase binding free energy (Gbinding 4.8 kcal molÿ1), although this may be due, in part, to structural changes induced by the mutation, since the conformational stability of the Tyr54Ala mutant decreased by 2.9 kcal mol ÿ1 (Wallis et al., 1998).

—-1emv-10.1006@jmbi.2000.3945.pdf

[conformational]

-- Use this when the amino acid is described as having a role in twisting or moving something into a different position because of flexibility or motion. Sometimes this co-occurs with steric interactions. Example, regarding Pro47:

Two notable exceptions, however, are Pro47 and Tyr54, which have significant portions of their side chains exposed to solvent but are very destabilizing when mutated to alanine (ΔΔGstability) 2.9 kcal/mol), reflecting their important structural roles. ***Pro47 is part of the turn between helices II and III which helps sculpt the interactions around Trp74****, and the side chain of Tyr54 sits in a pocket made up of Ile53 and Ser50 of helix III and Val37 of helix II. The crystal structure of Im7 shows the equivalent tyrosine residue forming a hydrogen bond with the base of helix I, thereby tethering the N- and C-terminal halves of the protein (12). This interaction is not apparent in the current solution structure of Im9, but this may be a consequence of insufficient distance restraints.*

—-1emv-10.1006@jmbi.2000.3945.pdf

[steric]

-- When the physical size or presence of the sidechain or backbone contributes to or detracts from binding, steric should be used. Example, from 1AK4:

"The CA151/CypA co-crystal structure indicated that CA Pro90 should be a major determinant of CypA binding, because the proline ring is buried in a deep hydrophobic pocket formed by CypA residues Phe60, Met61, Phe113 and Leu122. Analogous binding of proline residues has been observed in a number of cyclophilin A peptide complexes (Kallen & Walkinshaw, 1992; Ke et al., 1993; Zhao & Ke, 1996a,b). Detailed examination of the structure suggested that an Ala90 side-chain could be accommodated in the CypA

binding pocket, albeit with the loss of favorable binding interactions at the Pro90 Cg and Cd positions, whereas any larger side-chain should clash with the sides of the proline binding pocket. For example, a b-branched side-chain at CA residue 90 is predicted to clash with the CypA Leu122 sidechain. Consistent with this analysis, mutation of CA Pro90 to Ala reduced the CypA binding affinity by 2.1 kcal/mol, while mutation of Pro90 to Val resulted in immeasurably low (>3 kcal/mol reduced) CypA binding affinity."

--1ak4-000.pdf

[vanDerWaals]

-- Use this when the paper says that an amino acid makes VanDerWaals contacts with other amino acids. Sometimes they are vague and just say "contacts" with other amino acids. In such cases you need to make sure they are not referring to hydrogen bonds, but being this vague is rare. Example:

The UbcH7 Leu33, which occurs in a loop (residues 28 to 33) adjacent to the HI helix, makes a van der Waals contact to the E6AP Pro668.

–1c4z-huang1999.pdf

There are more, but the above are the most common.

I'm still working on adding others and making them standard in what I've already done..

2. G250 Mutation–PDB Index

The following table lists the 250 records in the supplied G250 index, in the original order. A row records the mutation and chain alongside its associated PDB structure identifiers. Repeated entries have been retained.

TABLE I
G250 Mutation–PDB Index

Mutation	PDB Structure(s)
K125A (chain P)	2BTF
P14Q (chain I)	1PPF; 1R0R
T484E (chain B)	5CXB
K28A (chain C)	1FCC
P14Y (chain I)	1PPF; 1R0R
P14V (chain I)	1PPF; 1R0R
R17A (chain I)	1CBW
T2R (chain D)	2J0T
Y437A (chain A)	1A4Y
K152R (chain B)	3SE4
Y437F (chain A)	1A4Y
P14V (chain I)	1PPF; 1R0R
P14I (chain I)	1PPF; 1R0R
P14K (chain I)	1PPF; 1R0R
K6A (chain B)	1XD3
A33I (chain B)	1XGP; 1XGQ; 1XGR; 1XGT; 1XGU
A33F (chain B)	1XGP; 1XGQ; 1XGR; 1XGT; 1XGU
A33I (chain B)	1XGP; 1XGQ; 1XGR; 1XGT; 1XGU
A33V (chain B)	1XGP; 1XGQ; 1XGR; 1XGT; 1XGU
E401A (chain A)	1A4Y
A15T (chain I)	1PPF; 1R0R
E486R (chain B)	5CYK
R67C (chain I)	1TM1
H29K (chain B)	1FSS
D52A (chain B)	1JTD; 1JTG
R21E (chain I)	1PPF
A15L (chain I)	1PPF; 1R0R
A15V (chain I)	1PPF; 1R0R
A15K (chain I)	1PPF; 1R0R
A15W (chain I)	1PPF; 1R0R
R21F (chain I)	1PPF
Y50A (chain B)	1JTD; 1JTG
D52A (chain B)	1JTD; 1JTG
D104E (chain A)	2G2U
R38T (chain A)	2WPT
R45K (chain I)	1SBN
T6K (chain P)	1KNE
R59A (chain A)	1BRS
K6R (chain B)	1XD3
E35Q (chain A)	2PCB; 2PCC
E35Q (chain A)	2PCB; 2PCC
S417P (chain A)	1P69; 1P6A
F47A (chain C)	3HH2
L18N (chain I)	1CHO
L18D (chain I)	1PPF
Y20E (chain I)	1CHO
Y50K (chain A)	1VFB
F176A (chain A)	1BP3
L18R (chain I)	1PPF; 1R0R
P13A (chain I)	1CBW
L18E (chain I)	1PPF; 1R0R
L18T (chain I)	1PPF
L18V (chain I)	1CHO
K13T (chain I)	1R0R
L18T (chain I)	1R0R
E41A (chain A)	2WPT
L33F (chain B)	1XGP; 1XGQ; 1XGR; 1XGT; 1XGU
Y58A (chain H)	1JRH
F47E (chain C)	2B0U
R21V (chain I)	1PPF
E78A (chain C)	3AAA
L18Q (chain I)	1CHO
T17V (chain I)	1PPF; 1R0R
W52A (chain B)	1DVF
E19C (chain I)	1R0R
V80A (chain C)	3SE4
L18C (chain I)	1CHO
T181A (chain A)	3SE3; 3SE4
V58A (chain B)	1E50
E98A (chain B)	1DVF
W44F (chain D)	1BRS
L18I (chain I)	1PPF
N36G (chain I)	1CHO
F47E (chain C)	2B0U
L18P (chain I)	1R0R
L41W (chain C)	4CVW
M71A (chain A)	3F1S
Y20R (chain I)	1CHO
N92A (chain A)	1MLC
L18E (chain I)	1PPF
T17L (chain I)	1PPF; 1R0R
L10A (chain B)	3EQS
K84E (chain B)	1C1Y
V159A (chain A)	1E50
N34V (chain A)	2WPT
V43A (chain B)	5M2O
F97A (chain A)	1EAW
D33L (chain A)	2WPT
L18C (chain I)	1PPF
L18G (chain I)	1PPF
L18H (chain I)	1PPF; 1R0R
E31A (chain B)	2G2U
L18F (chain I)	1PPF
F33L (chain B)	1XGP; 1XGQ; 1XGR; 1XGT; 1XGU
S182A (chain A)	3SE3
R21Y (chain I)	1PPF
R27Y (chain B)	1FSS
Q86A (chain C)	1REW
L18K (chain I)	1PPF
T44A (chain C)	3SE4
L18R (chain I)	1PPF; 1R0R
I103A (chain C)	3SE4
T87A (chain B)	1EMV
D100A (chain B)	1DVF
G141A (chain B)	2G2U
L18I (chain I)	1R0R
R41A (chain A)	2DVW

Mutation	PDB Structure(s)
Y60G (chain A)	1EAW
W375A (chain A)	1A4Y
T17A (chain I)	1PPF; 1R0R
Q155A (chain A)	2BNR
Y20D (chain I)	1CHO
L33F (chain B)	1XGP; 1XGQ; 1XGR; 1XGT; 1XGU
W7A (chain B)	3EQS
E19K (chain I)	1R0R
K32R (chain B)	1FSS
R21I (chain I)	1CHO
F47E (chain C)	2B0U
T58P (chain I)	1TM1
T161A (chain A)	1E50
K116A (chain C)	1VFB
L18W (chain I)	1R0R
Y134A (chain G)	4N8V
K13H (chain I)	1R0R
F101Y (chain B)	1KIP; 1KIQ; 1KIR
A92V (chain C)	4RA0
L43A (chain M)	4B0M
Y143A (chain B)	2G2U
T17F (chain I)	1PPF; 1R0R
K111A (chain A)	1JTD; 1JTG
R21I (chain I)	1PPF
R227E (chain A)	3WWN
R21C (chain I)	1PPF
M129A (chain A)	1JTD; 1JTG
D112K (chain A)	4RS1
L18F (chain I)	1CHO
T17M (chain I)	1PPF; 1R0R
V37A (chain A)	1EMV
R21P (chain I)	1CHO
Y32A (chain H)	1JRH
D71A (chain A)	3UIG; 3UII
D51A (chain A)	1EMV
T17D (chain I)	1PPF; 1R0R
Y20K (chain I)	1R0R
D33A (chain A)	1K8R
F33A (chain B)	1XGP; 1XGQ; 1XGR; 1XGT; 1XGU
E77A (chain C)	3SE4
F3A (chain B)	3EQS
Y87I (chain C)	3BX1
L18H (chain I)	1PPF
T17I (chain I)	1PPF; 1R0R
N36H (chain I)	1CHO
L18A (chain I)	1PPF
L18Q (chain I)	1PPF; 1R0R
I51E (chain C)	2B0U
M33F (chain B)	1FSS
H29K (chain B)	1FSS
Y240A (chain A)	3F1S
L79A (chain A)	5M2O
E19P (chain I)	1R0R
W63A (chain C)	1DQJ
K96E (chain D)	1C4Z
W263A (chain A)	1A4Y
R260E (chain A)	3AAA
R5A (chain B)	1A4Y
Y20A (chain I)	1CHO

Mutation	PDB Structure(s)
L18R (chain I)	1PPF
D38A (chain A)	1K8R
D52A (chain B)	1JTD; 1JTG
R17A (chain I)	1CBW
K74A (chain B)	2G2U
Y20M (chain I)	1CHO
L18K (chain I)	1PPF
N79A (chain I)	1JRH
L18S (chain I)	1CHO
K13S (chain I)	1R0R
K195A (chain B)	4RS1
L18D (chain I)	1R0R
I2A (chain C)	3BK3
L18F (chain I)	1PPF; 1R0R
F33L (chain B)	1XGP; 1XGQ; 1XGR; 1XGT; 1XGU
K173A (chain E)	4X4M
L18W (chain I)	1PPF; 1R0R
R227A (chain A)	4FZA
E60S (chain I)	1TM1
R21S (chain I)	1PPF
L18G (chain I)	1CHO
K74A (chain B)	1JTD; 1JTG
L18A (chain I)	1PPF
Y164A (chain A)	1BP3
N19A (chain C)	1VFB
P6A (chain B)	5XCO
T17L (chain I)	1PPF; 1R0R
N36I (chain I)	1CHO
T17R (chain I)	1PPF; 1R0R
E76A (chain D)	1BRS
L18G (chain I)	1R0R
E19W (chain I)	1R0R
R21K (chain I)	1PPF
D38N (chain A)	1K8R
F25A (chain A)	1BP3
G54A (chain H)	3SE9
M129A (chain A)	1JTD; 1JTG
L18T (chain I)	1PPF; 1R0R
L18F (chain I)	1PPF
K111N (chain A)	1JTD; 1JTG
L18H (chain I)	1CHO
L18S (chain I)	1PPF
W318A (chain A)	1A4Y
L18E (chain I)	1PPF
L18F (chain I)	1R0R
Q86A (chain C)	1REW
E73Y (chain A)	1BRS
D32A (chain B)	1DQJ
L33A (chain B)	1XGP; 1XGQ; 1XGR; 1XGT; 1XGU
N50A (chain B)	1JTD; 1JTG
K111N (chain A)	1JTD; 1JTG
V33A (chain B)	1XGP; 1XGQ; 1XGR; 1XGT; 1XGU
V552A (chain D)	4HRN
L18F (chain I)	1PPF; 1R0R
T17Y (chain I)	1PPF; 1R0R
R5A (chain D)	1C4Z
Y20Q (chain I)	1R0R
E73Q (chain A)	1BRS
L18N (chain I)	1PPF

Mutation	PDB Structure(s)
K97A (chain C)	1DQJ
T88V (chain C)	3BX1
F33I (chain B)	1XGP; 1XGQ; 1XGR; 1XGT; 1XGU
R21N (chain I)	1PPF
L30K (chain B)	4BFI
N36L (chain I)	1CHO
W263A (chain A)	1A4Y
T17N (chain I)	1PPF; 1R0R
Y20I (chain I)	1CHO
L18N (chain I)	1PPF; 1R0R
Q12A (chain B)	1A4Y
L18W (chain I)	1PPF
T17K (chain I)	1PPF; 1R0R
D33L (chain A)	2WPT
K62A (chain A)	3UIG; 3UII
S11A (chain B)	3EQS
Y240A (chain A)	3F1S
E73A (chain B)	1JTD; 1JTG
Y2F (chain G)	4E6K
R21C (chain I)	1CHO
R6A (chain D)	1C4Z
F178E (chain A)	4FZA
R21Q (chain I)	1CHO
D33L (chain A)	2WPT
K64E (chain D)	1C4Z